NLP-Based Candidate Recommendation System

Prepared by: Pasindu Edirisingha

Data Science Intern

Gamage Recruiters Pvt (Ltd)

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Submitted to: Jayani Bandara (Chief Executive Officer, Gamage Recruiters)

Team Members And Responsibilities

Member	Responsibility
Ahmed	Data Collection & Preprocessing
Huvini	Vectorization & Feature Engineering
Uththara	Similarity Computation & Recommendation
Vidarshana	Evaluation & Analysis
Pasindu	Documentation Report
Himandi	Presentation & Demo

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1. Introduction

Recruitment teams often receive multiple CVs for a single vacancy, making manual screening time-consuming and potentially inconsistent. This project develops an automated Candidate Recommendation System that uses Natural Language Processing (NLP) to compare candidate CVs with job descriptions and rank the most relevant candidates.

The system preprocesses unstructured CV and job-description text, converts the cleaned documents into numerical vector representations, computes candidate-to-job similarity, and produces Top-3 and Top-5 ranked recommendations. TF-IDF is used as the primary representation and Word2Vec is included as a comparison method. The final recommendation component uses Cosine Similarity.

The purpose of the system is to support the initial recruitment screening process. It does not replace recruiter judgement or make a final hiring decision; instead, it provides a repeatable ranking that helps recruiters focus attention on candidates whose documents show stronger textual relevance to a vacancy.

2. Project Objectives

- Preprocess and clean unstructured CV and job-description text using NLP techniques.
- Create candidate and job vector representations using TF-IDF and compare representation characteristics with Word2Vec.
- Investigate TF-IDF feature-engineering settings including n-grams and document-frequency thresholds.
- Calculate candidate-to-job match scores using Cosine Similarity.
- Generate ranked Top-3 and Top-5 candidate recommendations for each job opening.
- Evaluate recommendation quality using Precision, Recall and F1-Score against approved relevance labels and dataset ground truth.
- Document the complete workflow, implementation, findings, limitations and future improvements.

3. Dataset and Project Structure

3.1 Dataset Overview

The working corpus used by the implemented recommendation pipeline contains 12 cleaned candidate documents and 4 cleaned job descriptions. The candidate resume data originates from the Resume Dataset on Kaggle by Saugata Roy Arghya. A ground-truth file is maintained separately in the project repository for the evaluation stage.

Item	Detail
Candidate documents	12
Job descriptions	4
Total corpus	16 documents
Primary representation	TF-IDF
Comparison representation	Word2Vec
Matching method	Cosine Similarity
Evaluation cutoffs	Top-3 and Top-5

3.2 Repository Structure

Folder / File	Purpose
data/	Raw/cleaned CVs, job descriptions and ground truth
models/	Saved vectorizer and vector representations
src/	Preprocessing, vectorization, similarity and evaluation code
results/	Rankings, similarity scores and evaluation metrics
docs/	Project documentation and presentation artifacts

4. Methodology

The project follows a structured NLP recommendation methodology in which the output of each stage becomes the input to the next stage.

Stage 1 - Data Collection and Preprocessing: Candidate CVs and job descriptions are collected and standardized through text cleaning, tokenization, stop-word removal and lemmatization.

Stage 2 - Vectorization: Cleaned text is converted into numerical vectors. TF-IDF is the main representation; Word2Vec is used to compare vector characteristics.

Stage 3 - Feature Engineering: Alternative TF-IDF settings are compared to select a useful vocabulary size and representation for the small corpus.

Stage 4 - Similarity Computation: Each job vector is compared with all candidate vectors using Cosine Similarity.

Stage 5 - Recommendation: Candidates are sorted by similarity score and Top-3/Top-5 recommendations are produced.

Stage 6 - Evaluation: Rankings are compared with team-approved answer-key labels and dataset ground truth using Precision, Recall and F1-Score.

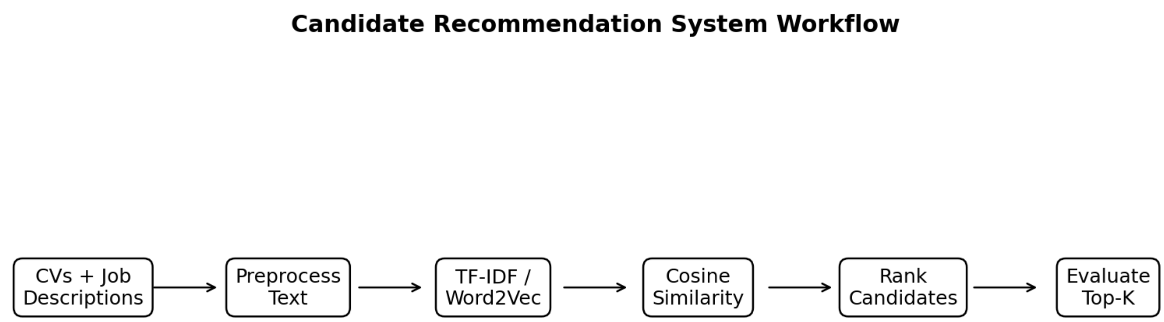

Figure 1: Overall Candidate Recommendation System Workflow

5. Data Collection and Preprocessing

Step 1: Collect Candidate CVs and Job Descriptions

Candidate CVs and job descriptions are the main inputs to the recommendation engine. Because both are unstructured text documents, the system first prepares them for consistent NLP processing.

Step 2: Text Cleaning

Text cleaning reduces formatting noise and standardizes the content before feature extraction. This helps the vectorization stage focus on skills, qualifications, job roles, experience and domain terminology rather than punctuation or inconsistent capitalization.

Step 3: Tokenization, Stop-Word Removal and Lemmatization

The project specification includes tokenization, stop-word removal and lemmatization. Tokenization separates text into processable units, stop-word removal reduces common low-information words, and lemmatization reduces related word forms to a more consistent base form.

Preprocessing Output

The cleaned candidate and job documents are stored separately so later vectorization and similarity components can work on standardized text without changing the original source documents.

6. Vectorization and Feature Engineering

6.1 TF-IDF Vectorization

TF-IDF (Term Frequency-Inverse Document Frequency) is the primary text representation. It gives higher weight to terms that are important in a document while reducing the influence of terms that appear across most documents. Candidates and jobs are transformed in a shared feature space so their vectors can be compared directly.

Parameter	Final Value	Purpose
ngram_range	(1, 2)	Uses individual words and two-word phrases
min_df	1	Retains rare candidate-specific skills in the small corpus
max_df	0.95	Removes terms appearing in almost every document
sublinear_tf	True	Reduces the dominance of repeated terms
Vocabulary	955 features	Shared candidate/job feature space

The final candidate matrix has shape (12, 955) and the job matrix has shape (4, 955).

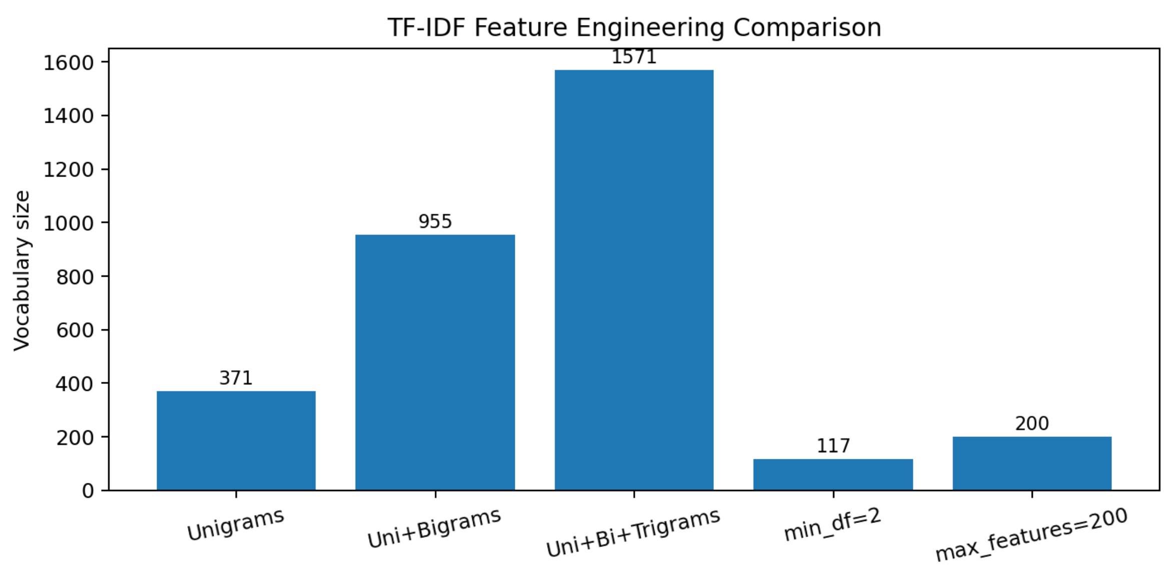

Figure 2: TF-IDF Feature Engineering Configuration Comparison

6.2 Feature Engineering Analysis

Five TF-IDF configurations were compared. Unigrams alone produced 371 features. Adding bigrams increased the vocabulary to 955 features, while adding trigrams expanded it to 1,571. Using min_df=2 reduced the vocabulary to 117 features, which could remove rare but useful candidate-specific skills. A max_features=200 cap also reduced the available information.

Configuration	Vocabulary	Density
Unigrams only	371	0.089
Unigrams + bigrams (selected)	955	0.075
Unigrams + bigrams + trigrams	1,571	0.070
min_df = 2	117	0.162
max_features = 200	200	0.121

Therefore, unigrams + bigrams with min_df=1, max_df=0.95 and sublinear_tf=True were selected. This preserves useful phrases such as domain skills while avoiding the larger feature space created by trigrams.

6.3 TF-IDF vs Word2Vec

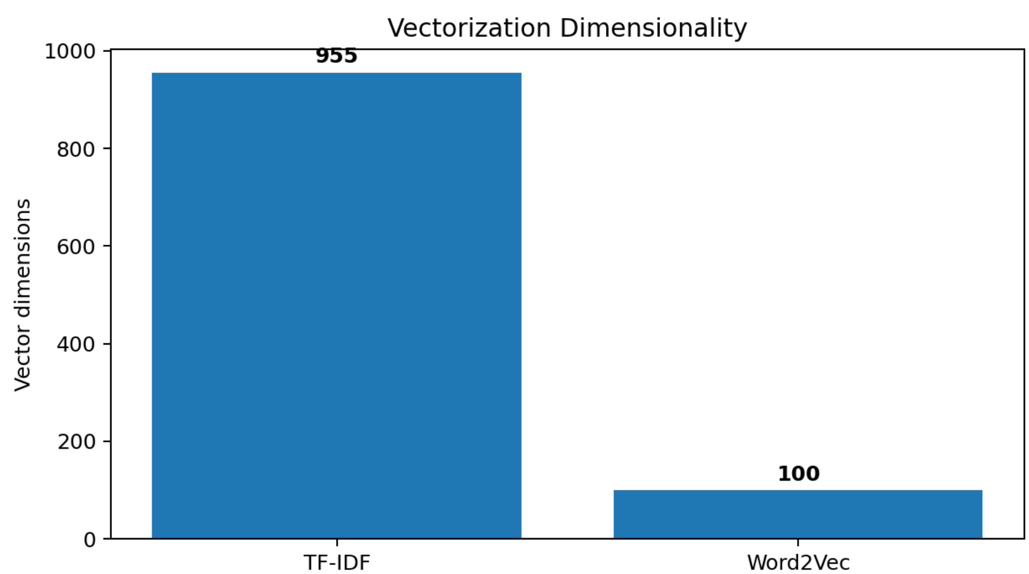

Figure 3: TF-IDF and Word2Vec Vector Dimensionality

Characteristic	TF-IDF	Word2Vec
Dimensions	955	100
Representation	Sparse	Dense
Non-zero density	0.080	1.000
Interpretability	High	Lower
Use in current system	Primary	Comparison

Word2Vec can capture semantic relationships, but the project contains only 12 CVs and 4 job descriptions. Because this is a very small training corpus, Word2Vec is treated as a comparison method rather than the primary representation. A pretrained sentence-embedding model is a suitable future improvement.

7. Similarity Computation and Recommendation

7.1 Cosine Similarity

Cosine Similarity measures the angle between two document vectors. A larger value indicates stronger similarity in the shared vector space. The recommendation component compares every job with all 12 candidates, producing 48 job-candidate similarity pairs.

$$\textbf{Cosine Similarity}(A, B) = (A \cdot B) / (||A||\ ||B||)$$

7.2 Ranking Process

- Load the saved candidate and job TF-IDF vectors.
- Calculate the cosine similarity matrix between 4 jobs and 12 candidates.
- Sort each job's candidate scores in descending order.
- Assign rank numbers.
- Export complete rankings, similarity scores and Top-5 recommendations.

The final similarity scores range from 0.0000 to 0.0660, with a mean of 0.0192. These values are similarity scores rather than hiring probabilities or percentages.

8. Candidate Recommendation Results

The recommendation component produces Top-5 candidate rankings for each of the four job descriptions. The following sections summarize the actual repository outputs.

8.1 job_01 - Senior Software Engineer

Rank	Candidate	Similarity Score
1	candidate_05	0.0522
2	candidate_03	0.0441
3	candidate_07	0.0316
4	candidate_11	0.0274
5	candidate_01	0.0258

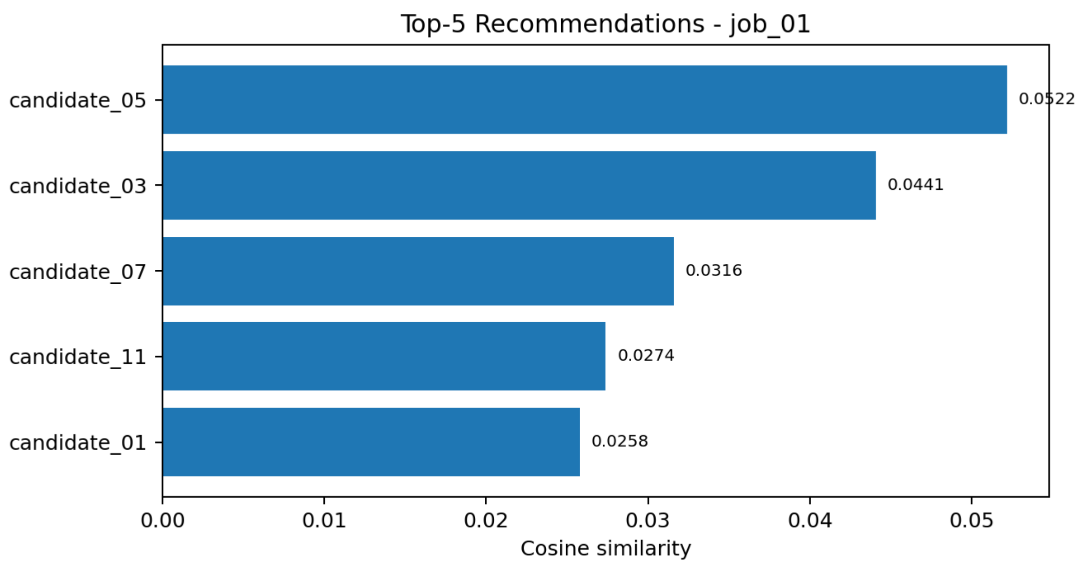

Figure 4: Top-5 Candidate Recommendations for job_01

candidate_05 is ranked first with a score of 0.0522, followed by candidate_03 at 0.0441. High-weight job terms include software, monitoring, field visit, industry and software engineer.

8.2 job_02 - HR Officer

Rank	Candidate	Similarity Score
1	candidate_08	0.0660
2	candidate_12	0.0487
3	candidate_07	0.0407
4	candidate_10	0.0264
5	candidate_11	0.0230

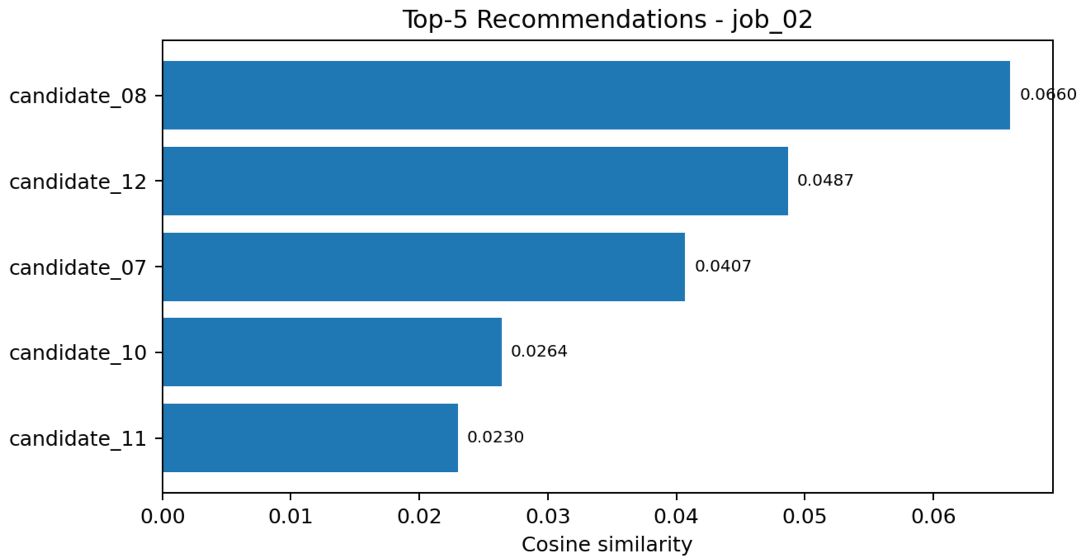

Figure 5: Top-5 Candidate Recommendations for job_02

candidate_08 receives the highest raw similarity score (0.0660), while candidate_12 is second. Important job terms include resource management, HR, management, policy and human resource.

8.3 job_03 - Civil Engineer

Rank	Candidate	Similarity Score
1	candidate_06	0.0543
2	candidate_07	0.0399
3	candidate_03	0.0316
4	candidate_05	0.0243
5	candidate_02	0.0226

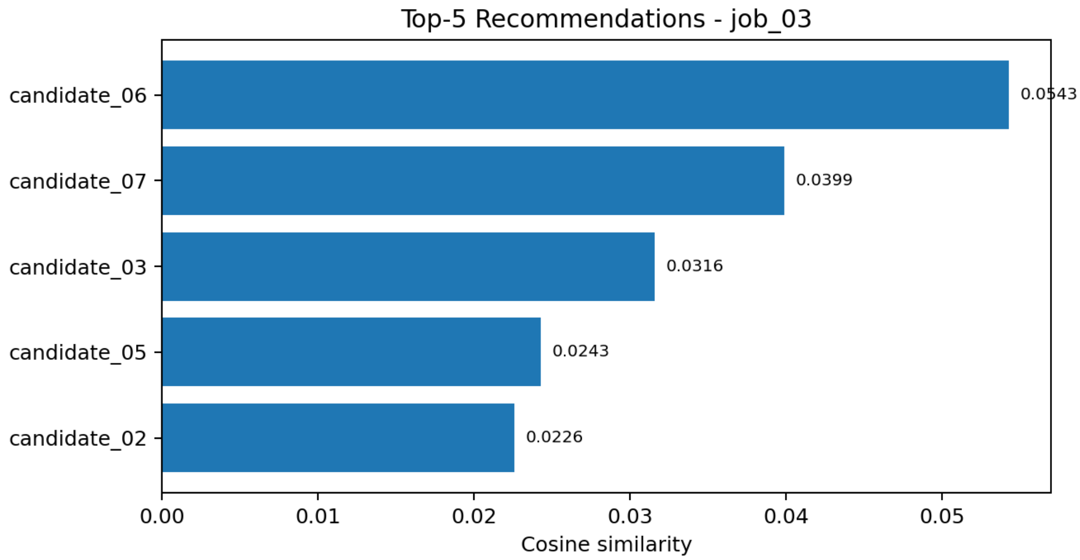

Figure 6: Top-5 Candidate Recommendations for job_03

candidate_06 is ranked first with a score of 0.0543. The job representation emphasizes review, civil, utilization and feasibility terminology.

8.4 job_04 - Business Development Executive

Rank	Candidate	Similarity Score
1	candidate_08	0.0658
2	candidate_11	0.0246
3	candidate_10	0.0153
4	candidate_01	0.0118
5	candidate_04	0.0094

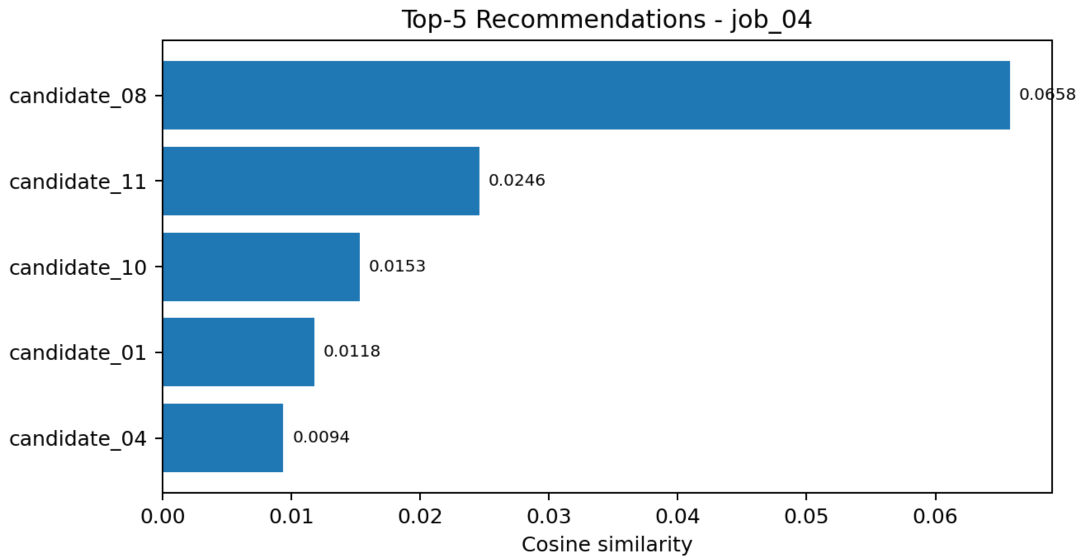

Figure 7: Top-5 Candidate Recommendations for job_04

candidate_08 is ranked first with a score of 0.0658. The job representation strongly emphasizes sourcing and also includes partner, vendor communication and quality terminology.

9. Evaluation and Analysis

9.1 Evaluation Setup

The final team-approved answer-key evaluation contains 48 job-candidate pairs across 4 jobs, with 8 pairs labelled relevant. Precision@K, Recall@K and F1@K are calculated at Top-3 and Top-5 cutoffs. Macro averages give equal weight to each job, while micro averages aggregate true positives, false positives and false negatives across all jobs.

Job	Relevant Candidates
job_01 - Senior Software Engineer	3
job_02 - HR Officer	1
job_03 - Civil Engineer	1
job_04 - Business Development Executive	3
Total relevant pairs	8

9.2 Top-3 Results

Job	TP	FP	FN	Precision	Recall	F1
job_01	2	1	1	0.6667	0.6667	0.6667
job_02	1	2	0	0.3333	1.0000	0.5000
job_03	1	2	0	0.3333	1.0000	0.5000
job_04	2	1	1	0.6667	0.6667	0.6667

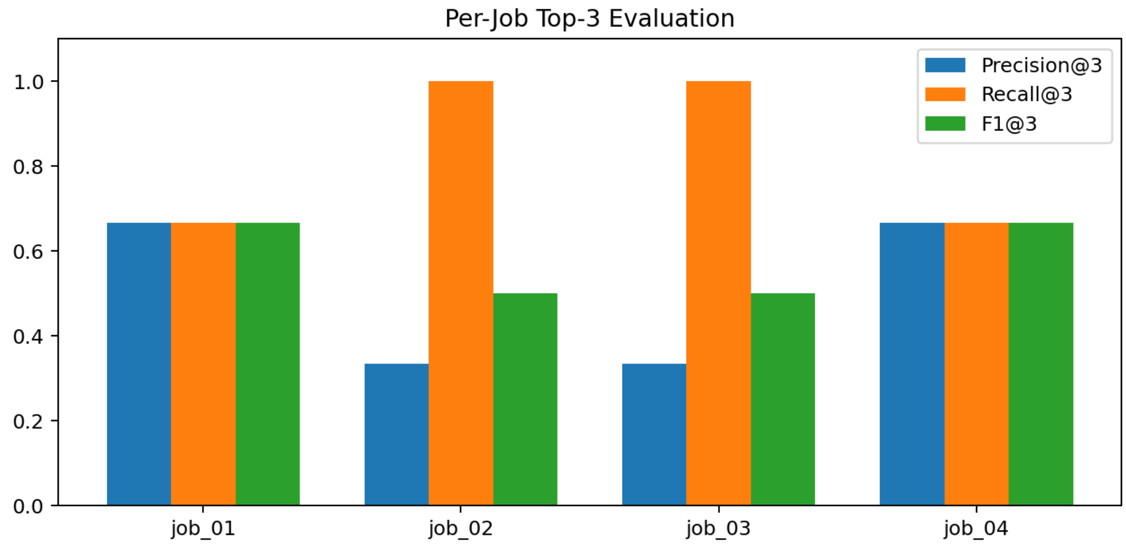

Figure 8: Per-Job Top-3 Precision, Recall and F1

9.3 Overall Top-3 and Top-5 Performance

Cutoff	Average	Precision	Recall	F1
Top-3	Macro	0.5000	0.8333	0.5833
Top-3	Micro	0.5000	0.7500	0.6000
Top-5	Macro	0.3000	0.8333	0.4167
Top-5	Micro	0.3000	0.7500	0.4286

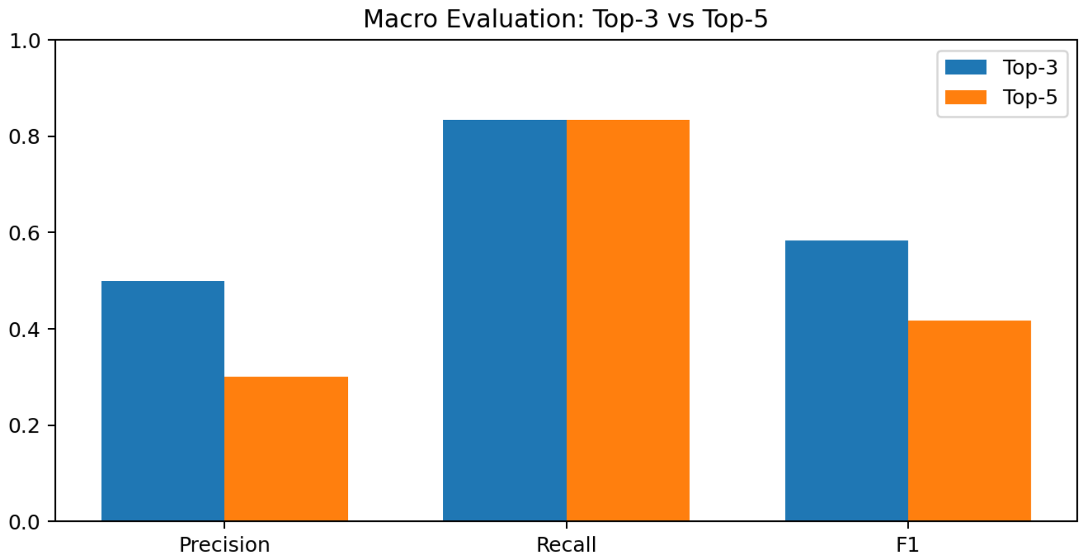

Figure 9: Macro Evaluation Comparison - Top-3 vs Top-5

The Top-3 recommendations retrieved 6 of the 8 relevant pairs overall. Expanding the list from Top-3 to Top-5 added no additional relevant candidates. Therefore, Top-3 is the more efficient cutoff for the current dataset: recall remains unchanged while precision and F1 are higher than at Top-5.

9.4 Relevant Candidate Ranks

Job	Relevant Candidate Ranks
job_01	candidate_05 (1), candidate_03 (2), candidate_02 (6)
job_02	candidate_12 (2)
job_03	candidate_06 (1)
job_04	candidate_08 (1), candidate_10 (3), candidate_12 (10)

9.5 Dataset Ground-Truth Evaluation

A second evaluation compares the TF-IDF rankings with data/ground_truth.csv. The file contains matched scores for 12 of the 48 possible pairs; the other 36 pairs are assumed non-relevant for Top-K classification. A matched_score threshold of 0.50 is used to derive relevance labels.

Metric	Value
Observed ground-truth pairs	12
All ranked pairs	48
Relevant pairs at threshold	9
Ground-truth mean score	0.5986
Cosine-similarity mean	0.0183
MAE	0.5803
RMSE	0.6052
Pearson correlation	0.3716
Spearman correlation	0.3099

For this ground-truth evaluation, Top-3 macro Precision is 0.2500, Recall is 0.4583 and F1 is 0.3083. The lower agreement is consistent with sparse exact-term TF-IDF matching and the fact that cosine similarity is not calibrated to reproduce the continuous matched-score scale.

9.6 Actual Evaluation Output

```
Candidate Recommendation Evaluation & Analysis
====================================================
Status: FINAL - answer key approved by the project team

Evaluation setup
----------------
Recommendation method: TF-IDF cosine similarity
Jobs: 4
Candidates per job: 12
Answer-key pairs: 48
Relevant pairs: 8
Cutoffs: Top-3, Top-5

Relevant candidates by job:
  job_01 (Senior Software Engineer): 3
  job_02 (HR Officer): 1
  job_03 (Civil Engineer): 1
  job_04 (Business Development Executive): 3

Metric definitions
------------------
Precision@K = relevant candidates in Top-K / K
Recall@K = relevant candidates in Top-K / all relevant candidates
F1@K = harmonic mean of Precision@K and Recall@K
Macro average = equal weight for each job
Micro average = aggregate TP/FP/FN across all jobs

Top-3 results
---------------
job_01 (Senior Software Engineer): TP=2, FP=1, FN=1, Precision=0.6667, Recall=0.6667, F1=0.6667
  Relevant recommended: candidate_03;candidate_05
  Missed relevant: candidate_02
job_02 (HR Officer): TP=1, FP=2, FN=0, Precision=0.3333, Recall=1.0000, F1=0.5000
  Relevant recommended: candidate_12
  Missed relevant: None
job_03 (Civil Engineer): TP=1, FP=2, FN=0, Precision=0.3333, Recall=1.0000, F1=0.5000
  Relevant recommended: candidate_06
  Missed relevant: None
job_04 (Business Development Executive): TP=2, FP=1, FN=1, Precision=0.6667, Recall=0.6667, F1=0.6667
  Relevant recommended: candidate_08;candidate_10
  Missed relevant: candidate_12

Macro: Precision=0.5000, Recall=0.8333, F1=0.5833
Micro: Precision=0.5000, Recall=0.7500, F1=0.6000

Top-5 results
---------------
job_01 (Senior Software Engineer): TP=2, FP=3, FN=1, Precision=0.4000, Recall=0.6667, F1=0.5000
  Relevant recommended: candidate_03;candidate_05
  Missed relevant: candidate_02
job_02 (HR Officer): TP=1, FP=4, FN=0, Precision=0.2000, Recall=1.0000, F1=0.3333
  Relevant recommended: candidate_12
  Missed relevant: None
job_03 (Civil Engineer): TP=1, FP=4, FN=0, Precision=0.2000, Recall=1.0000, F1=0.3333
  Relevant recommended: candidate_06
  Missed relevant: None
job_04 (Business Development Executive): TP=2, FP=3, FN=1, Precision=0.4000, Recall=0.6667, F1=0.5000
```

Figure 10: Final Evaluation Output from the Repository

10. Overall Findings

- The final TF-IDF representation uses 955 features generated from unigrams and bigrams.
- TF-IDF is a strong interpretable baseline for this small corpus; Word2Vec is limited by the small amount of training text.
- The system successfully generates ranked recommendations for all four job descriptions.
- The team-approved evaluation shows Top-3 retrieves 6 of 8 relevant pairs overall.
- Top-5 does not retrieve any additional relevant candidates, so Top-3 provides better precision and F1 for this dataset.
- Absolute cosine-similarity values are low because TF-IDF rewards exact term overlap and many CV/job pairs use different wording.
- Some relevant candidates appear outside the Top-5, showing that semantic matching remains an important improvement area.

11. Limitations and Ethical Considerations

- The candidate pool is small and contains only 12 CVs and 4 job descriptions.
- The number of relevant candidates is uneven across jobs; jobs with only one relevant candidate have a low maximum possible Precision@K.
- TF-IDF cannot reliably identify semantic matches when the same skill is expressed using different wording.
- The system does not separately weight skills, years of experience, education and certifications.
- The answer-key metrics depend on team-approved manual relevance judgements.
- Automated ranking should support rather than replace human recruitment decisions. Sensitive or protected characteristics should not be used to unfairly influence candidate ranking.

12. Conclusion

This project successfully developed an NLP-based Candidate Recommendation System that transforms CVs and job descriptions into numerical representations and ranks candidates according to textual similarity. The completed pipeline includes preprocessing, TF-IDF vectorization, Word2Vec comparison, feature engineering, cosine-similarity scoring, candidate ranking and quantitative evaluation.

The selected TF-IDF configuration uses unigrams and bigrams with min_df=1, max_df=0.95 and sublinear term frequency, producing a manageable 955-feature space. The final team-approved evaluation demonstrates that Top-3 recommendations are preferable to Top-5 for the current dataset because they retrieve the same relevant candidates with higher precision and F1.

The project provides a useful baseline for automated recruitment screening while also showing the limitations of exact-term matching. The strongest next step is to introduce pretrained semantic embeddings and structured skill/experience features, then evaluate the improved system on a larger and more diverse candidate dataset.

13. Recommendations and Future Improvements

1. Use pretrained Sentence-BERT or another sentence-embedding model to capture semantic similarity beyond exact word overlap.
2. Extract and separately weight mandatory skills, experience, education and certifications.
3. Increase the number and diversity of CVs and job descriptions used for testing.
4. Create a recruiter interface that displays ranked candidates together with explainable match factors.
5. Perform fairness and bias testing before using the system in real recruitment decisions.
6. Continue monitoring Precision, Recall and F1 as the dataset and relevance labels are expanded.

14. References

Roy Arghya, S. Resume Dataset. Kaggle. https://www.kaggle.com/datasets/saugataroyarghya/resume-dataset

Candidate Recommendation Engine GitHub repository - himandi02/candidate-recommendation-engine (project source code and results).